Statistics 652 - 2026

Homework Set 1

1.1 Define $g(\mu,\theta) = e^{-\mu^2\theta} + e^{-\theta} + \frac{1}{3}\log\theta$ on the convex set $C = \{(\mu,\theta) \mid 0 < \theta < \infty, \mu > \theta - 1\}$. Show that g has only one critical point in C, and that this critical point is a local maximum but not a global maximum.

1.2 Consider a random sample from a $Normal(\mu, \sigma^2)$ with $\mu \in \Re$ and $\sigma^2 > 0$.

i) Show the likelihood goes to zero as (μ, σ^2) goes to the boundary of the parameter space.

ii) Show the likelihood function is not strictly log concave in (μ, σ^2).

ii) Verify directly that the Hessian of the log likelihood in strictly concave in (η_1, η_2), where $\eta_1 = 1/\sigma^2$ and $\eta_2 = \mu/\sigma^2$.

1.3 i) Show that if $f(x;\theta)$ is an exponential family in $\theta = (\theta_1, \theta_2)$, then for fixed θ_2 it is an exponential family in θ_1.

ii) (Optional) For $\theta = (\theta_1, \theta_2)$, suppose, for fixed θ_2, $f(x;\theta)$ is an exponential family in θ_1, and, for fixed θ_1, $f(x;\theta)$ is an exponential family in θ_2. Is $f(x;\theta)$ an exponential family in θ?

1.4 (Optional) Show a k-dimensional exponential family is in minimal representation form if and only if

i) $\sum_{j=1}^k a_j c_j(\theta) = a_0$ for all $\theta \in \Theta \Leftrightarrow a_0 = \cdots = a_k = 0$.

ii) $\sum_{j=1}^k b_j T_j(x) = b_0$ for all $x \in A \Leftrightarrow b_0 = \cdots = b_k = 0$.

1.5 i) For $\mu \in \Re$, show that $f(x;\mu) = \frac{2}{\Gamma(1/4)} e^{-(x-\mu)^4}$, for $-\infty < x < \infty$, is a k-dimensional curved exponential family, and specify k.

ii) For a random sample of size n from the above distribution, how many roots are there to the likelihood equations? Which, if any of the roots, corresponds to the maximum likelihood estimate?

iii) Find a minimal sufficient statistic for μ.

1.6 Show that if $\boldsymbol{\eta}$ represents the natural parameters for an exponential family, then $\boldsymbol{\eta}_*$ is also a natural parametrization for the same exponential family if and only if $\boldsymbol{\eta}_* = \mathbf{A}\boldsymbol{\eta} + \mathbf{a}$ for some nonsingular $k \times k$ matrix $\mathbf{A}$ and some $\mathbf{a} \in \Re^k$.

1.7 Consider a k-dimensional exponential family in natural form.

i) Show that $e^{-d_o(\boldsymbol{\eta})}$ is a strictly convex function of $\boldsymbol{\eta}$.

ii) Show that $d_o(\boldsymbol{\eta})$ is a strictly concave function of $\boldsymbol{\eta}$.

iii) Show $\nabla d_o(\boldsymbol{\eta}) = -E_\eta[T(\mathbf{X})]$.

iv) Show $\frac{\partial^2 d_o(\boldsymbol{\eta})}{\partial\eta_i \partial\eta_j} = -Cov_\eta[T_i(\mathbf{X}), T_j(\mathbf{X})]$.

v) Let $\mathbf{H}$ represent the Hessian of $l(\eta) = \log L(\eta; x)$. Show

$$\mathbf{a}^T\mathbf{H}\mathbf{a} = \sum_{i=1}^k \sum_{j=1}^k a_i a_j h_{ij} = -Var_\eta\left(\sum_{j=1}^k a_j T_j(X)\right).$$

vi) Find the moment generating function of the natural sufficient statistic $T(\mathbf{X})$ for a k-dimensional exponential family.

1.8 Find the maximum likelihood estimates if they exist for the following cases.

i) $X_1 \sim Normal(\mu, \sigma^2)$, $X_2 \sim Normal(\mu, 1)$, with X_1 and X_2 independent and for $\mu \in \Re$ and $\sigma^2 > 0$.

ii) $X_1, \ldots, X_n$ i.i.d. $Normal(\theta, \theta^2)$ for $\theta \in \Re$.

iii) $X_1, \ldots, X_n$ i.i.d. $Normal(\theta, \theta)$ for $\theta > 0$.

iv) $X_1, \ldots, X_n$ i.i.d. $Gamma(\theta, \theta)$ for $\theta > 0$.

v) $X_1, \ldots, X_n$ i.i.d. $Gamma(\theta, 1/\theta)$ for $\theta > 0$.

Note: The density for the gamma is $f(x; \alpha, \beta) = \{\Gamma(\alpha)\beta^\alpha\}^{-1} x^{\alpha-1} e^{-x/\beta}$.

1.9 Suppose $X_1, \ldots, X_n$ are i.i.d. $Bernoulli(\theta)$. For a given integer k, find the UMVUE for θ^k. Also, under what conditions does the UMVUE for θ^k not exist?

1.10 Prove that any function of a complete statistic is also complete.

1.11 Suppose $X_1, \ldots, X_n$ are i.i.d. $Poisson(\lambda)$ for $\lambda > 0$.

i) Show there does not exist an unbiased estimate of $g(\lambda) = |\lambda - 1|$.

ii) For a given value of $k \in \Re$, find the UMVUE for $g_k(\lambda) = \lambda^k$ if it exists.

iii) For a given value of $a > 0$, find the UMVUE for $g_a(\lambda) = e^{-a\lambda}$ if it exists. Why might this estimate be undesirable when $n < a$, and in particular when $a = 2n$?

1.12 For a regular one parameter family $f(x; \theta)$, show that the relationship between the information for θ and the information for $\gamma = g(\theta)$, with g being $1-1$, is given by $I(\gamma; X) = I(\theta; X)/\{g'(\theta)\}^2$.

1.13 Prove that for a $k = 1$ dimensional exponential family in natural form, $I(\eta; X) = -d_o''(\eta)$. Also, in it original form, i.e. $\theta = c^{-1}(\eta)$, show $I(\theta; X) = -d''(\theta) + d'(\theta)c''(\theta)/c'(\theta)$.

1.14 Let $X_1, \ldots, X_n$ be i.i.d. $Normal(\theta, \theta)$ for $\theta > 0$.

i) Find the Fisher information $I(\theta; X_1, \ldots X_n)$.

ii) Both the sample mean $\bar{X}$ and the sample variance s^2 are unbiased estimators of θ. Find the Fisher efficiency of these two unbiased estimators, and indicate which has the better efficiency?

iii) Does there exist an unbiased estimator of θ which uniformly improves upon the Fisher efficiency of both $\bar{X}$ and s^2? If so, discuss how one would find such an estimator of θ. If not, justify why not.

iv) Does there exist an unbiased estimator of θ which achieves a Fisher efficiency of one?

1.15 Let $X_1, \ldots, X_n$ be i.i.d. $Poisson(\lambda)$ for $\lambda > 0$, and let $\hat{\gamma}$ represent the UMVUE for $\gamma = e^\lambda$.

i) Find the mean and variance of $\hat{\gamma}$.

ii) Show directly that $var(\hat{\gamma}) > 1/I(\gamma; X)$.

iii) Show $\frac{var(\hat{\gamma})}{1/I(\gamma;X)} \to 1$ as $n \to \infty$.

1.16 (Optional) Prove: If $T(X) \in \Re^k$ is a sufficient statistic for a general parametric model, and if there exists $\theta_0, \ldots, \theta_k \in \Theta$ such that

$$\left\{ \frac{f(x, \theta_1)}{f(x, \theta_0)}, \ldots, \frac{f(x, \theta_k)}{f(x, \theta_0)} \right\} \text{ is a 1-1 function of } T(x),$$

then $T(x)$ is minimal sufficient.

1.17 Suppose $X \sim$ Gamma(α, β), with $\alpha > 0,\ \beta > 0$. Using the exponential family representation for the gamma distribution, find

i) $E[log(X)]$, *ii)* VAR$[log(X)]$, and iii) COV$[log(X), X]$.

1.18 Let $\mathbf{X} = (X_1, \ldots, X_n)$ be i.i.d. Gamma(α, β), and let $g : \Re^n \to \Re$. Prove or disprove: If $E[g(\mathbf{X})] = 0$ for all $\alpha > 0,\ \beta > 0$, then $E\left[\ g(\mathbf{X}) \mid \sum_{i=1}^n X_i,\ \prod_{i=1}^n X_i\ \right] = 0$ for all $\alpha > 0,\ \beta > 0$.

1.19 Suppose $Y_1, \ldots, Y_n$ are independent with $Y_i \sim Normal(\beta x_i, \sigma^2)$, with $x_1, \ldots, x_n$ being fixed known constants. The parameters $\beta \in \Re$ and $\sigma^2 > 0$ are unknown.

i) Find complete sufficient statistics. *Clearly justify your answer.*

ii) Find the maximum likelihood estimators of (β, σ)

iii) Find the UMVUEs for (i) β, and (ii) σ^2